AAI-500 TEAM PROJECT

Predicting successful NFL third-down conversions

Which game and play conditions influence whether an offense converts?

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Project introduction · data processing · EDA

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Model Selection · Analysis · Conclusion

DATA PROCESSING

From 161,770 plays to 22,766 usable third downs

161,770
All play-by-play records
2023–2026 · 372 columns

→

26,343
Third-down records
Down = 3

→

22,766
Final analysis population
Regular season · run/pass · valid outcome

16 selected variables

0 missing selected values

0 duplicate plays

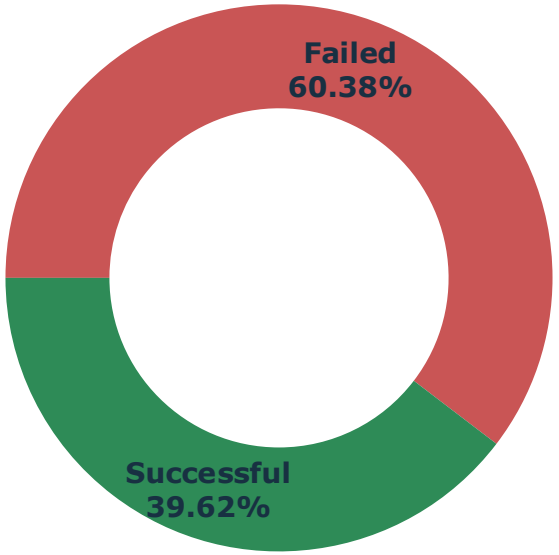
RESULT

The final dataset contains one clean row per regular-season third-down run or pass attempt.

EDA BASELINE

The overall third-down conversion rate was 39.62%

Third-down outcomes



9,021
successful

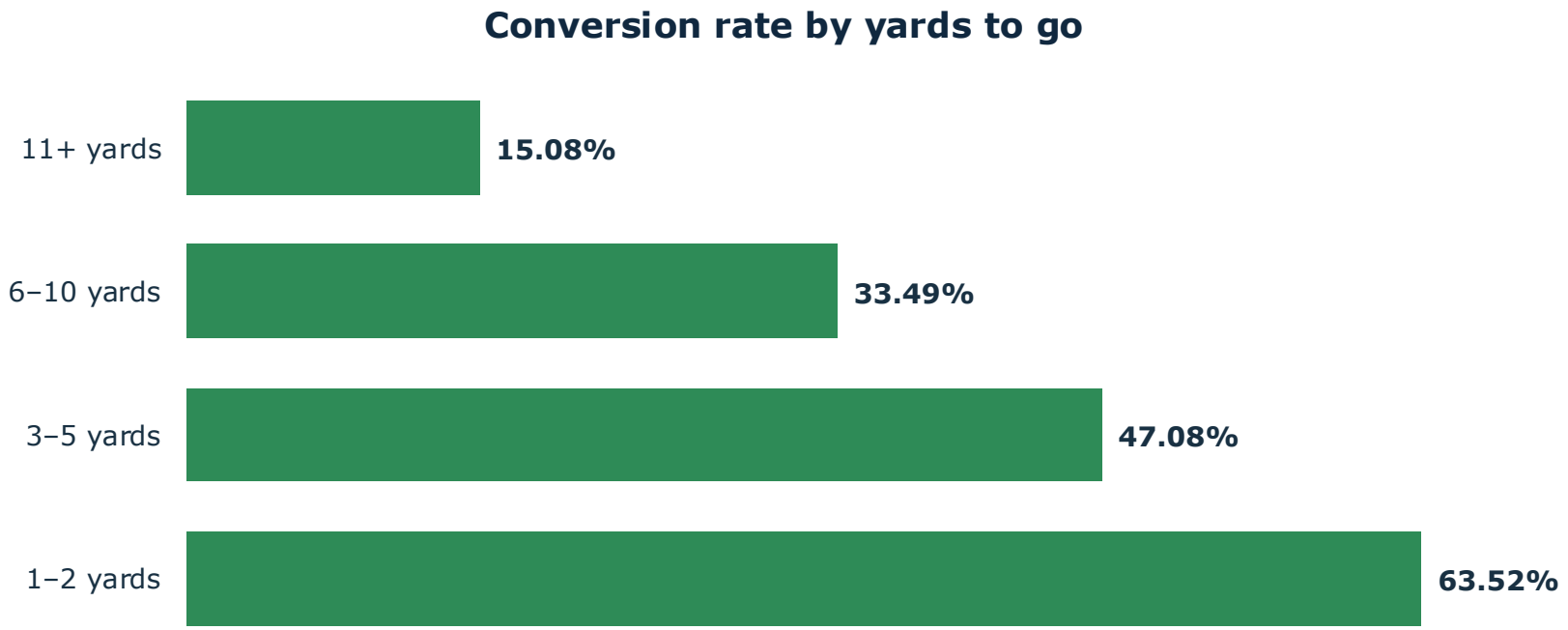
13,745
failed

BASELINE

A simple model must beat the majority-class baseline, not just report overall accuracy.

EDA FINDING 1

Distance had the clearest relationship with conversion success



63.52%
on 3rd-and-1 or 2

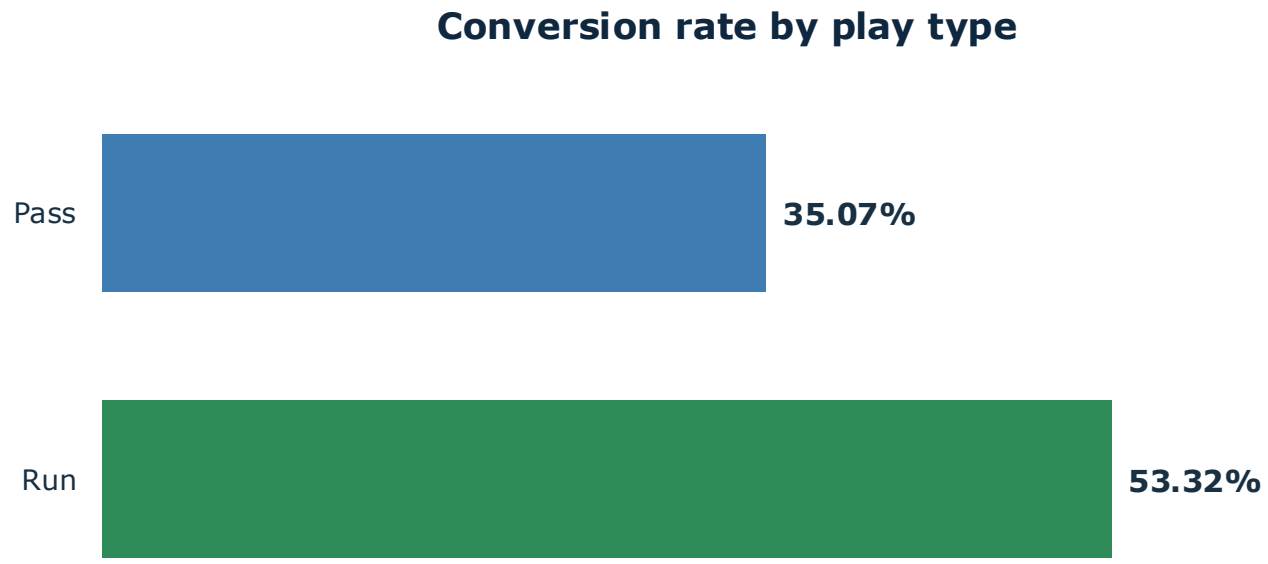
vs. 15.08% on 11+ yards

INTERPRETATION

Conversion probability dropped steadily as the required distance increased.

EDA FINDING 2

Run plays converted more often than pass plays



Important context

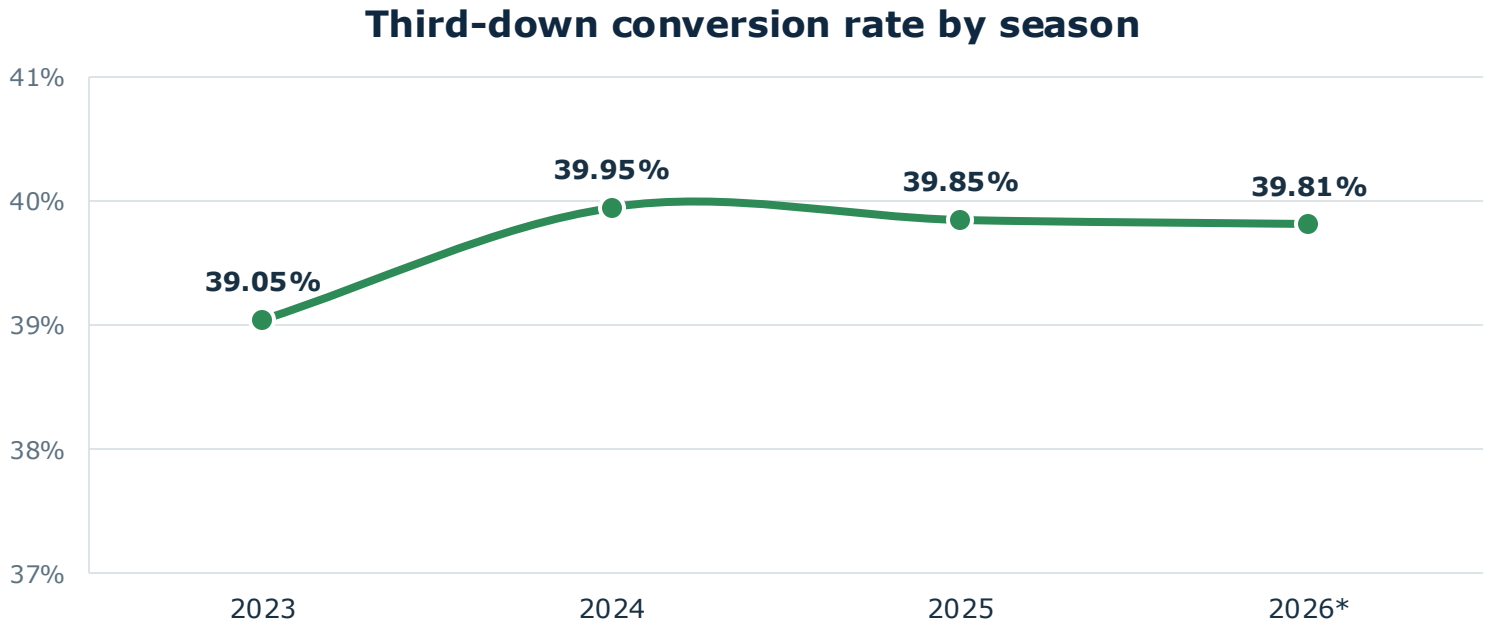
Teams usually choose runs in shorter-yardage situations. This comparison does not prove that running causes a higher success rate.

USE IN MODELING

Play type should be considered together with yards to go and other game conditions.

EDA FINDING 3 + HANDOFF

Conversion rates were stable across seasons



Other differences were smaller

Field position
30.50%–41.03%

Score situation
38.27%–40.83%

** 2026 is an incomplete season.*

HANDOFF

These patterns established the baseline and context for the modeling analysis.

Three models for a yes-or-no conversion

Same predictors: yards to go, called run versus pass, and goal-to-go. Fit on 20,769 plays, 2023–2025. Scrambles are coded as passes.

Linear

A straight line on the probability.

501 fitted chances fall outside 0 to 1.

The highest fitted chance is 0.66, below the 0.74 observed for a called run on open-field 3rd-and-1.

At 20 yards the called-run line is negative.

Identity link

The same straight line, with a yes-or-no variance.

Only 4 fitted chances leave 0 to 1.

At 1 yard it gives 0.61 and 0.45, against observed 0.74 and 0.58.

At 20 yards it is still near 0.23 to 0.28, against 0.00 and 0.03.

Logistic

The line is on the log-odds, so every chance stays between 0 and 1.

No fitted chance on these 20,769 plays leaves 0 to 1.

At 1 yard the fitted chances are 0.71 and 0.60.

At 20 yards they are 0.00 and 0.08.

We did not use AIC to choose. The linear model and the logit do not share a likelihood, and the identity-link fit does not return a usable AIC.

We kept the logistic model

It is the specification whose curve follows the drop in the observed rates.

Why this one

Every fitted probability on the 20,769 plays stays between 0 and 1.

A pass and a called run get different slopes.

The two lines cross at 3.7 yards in the open field.

Goal-to-go moves both lines by the same factor on the odds.

Pass

$$\log\left(\frac{\pi}{1-\pi}\right) = 0.538 - 0.150 \times (\text{yards to go})$$

Called run

$$\log\left(\frac{\pi}{1-\pi}\right) = 1.222 - 0.333 \times (\text{yards to go})$$

One more yard multiplies a pass's odds by 0.86.
The 95% interval is 0.85 to 0.87.

3.7 yards

Open-field crossover.
Called run is higher before it.

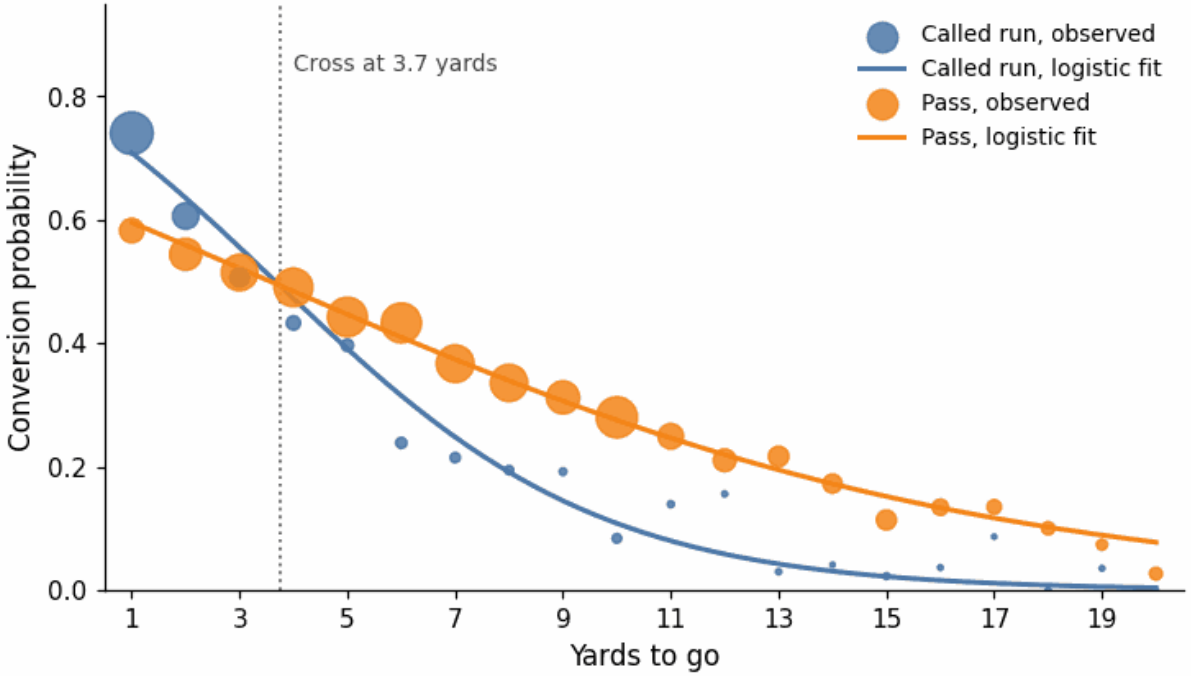
0.635

Goal-to-go multiplies the odds.
Interval: 0.56 to 0.72.

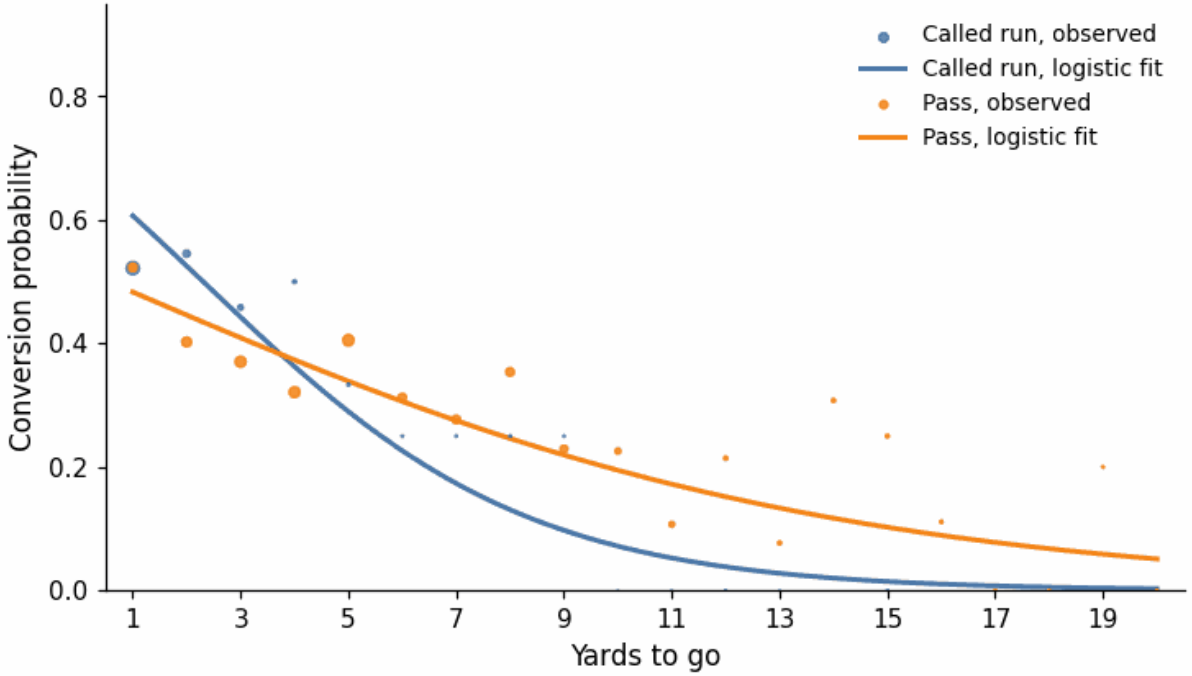
20,769

Plays used to estimate.
1,179 from 2026 held out.

The called run is higher only while the distance is short



Open field, 2023–2025



Goal line, 2023–2025

3rd-and-1

Open field

0.71 called run

0.60 pass

3rd-and-10

Open field

0.11 called run

0.28 pass

3rd-and-1

Goal line

0.61 called run

0.48 pass

What the model is for

These chances describe the calls that were made. They are not the effect of forcing the other call.

Inside about 4 yards

The called run is higher. On open-field 3rd-and-1 the fitted chances are 0.71 and 0.60.

Past that distance

The pass is higher. On open-field 3rd-and-10 they are 0.11 for a called run and 0.28 for a pass.

On the goal line

The odds are multiplied by 0.635. That 3rd-and-1 called run falls from 0.71 to 0.61.

A scramble stays a pass

The file labels it a run. Here it is not evidence for calling a run.

2026, held out

0.393

predicted rate

0.412

observed rate
1,179 plays
Weeks 1–3 only

Close to the 39.3% the earlier seasons implied. Too small a sample to refit the line.

Plays in the same game are dependent, so ordinary standard errors are somewhat too small. The gaps at 1 yard and at 10 yards are large next to that extra uncertainty.